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The Impact of Safety Culture, Risk Mitigation, and Sustainability on Business Performance: The Mediating Role of Employee Engagement in Palestinian Small and Medium-Sized Enterprises

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Abstract: Small and medium-sized enterprises (SMEs) in Palestine face significant challenges in risk management, in which safety culture, risk mitigation, and sustainability practices play a critical role in business performance. However, the ways these variables influence business performance through employee engagement are less explored. This study investigates the impact of safety culture, risk mitigation, and sustainability practices on the business performance of Palestinian SMEs, with a special emphasis on employee engagement's mediating effect. The quantitative approach was adopted, and data were collected from 450 SME respondents within the main clusters of the West Bank (Ramallah, Hebron, Nablus, and Jericho) based on purposive sampling. The findings indicate that safety culture, risk mitigation, and sustainability practices positively and significantly impact business performance. Furthermore, employee engagement is a potent mediator that enhances these variables' positive effects on business performance. These results have practical and theoretical implications for building a robust safety culture, possessing effective risk minimization measures, and incorporating sustainability practices to advance business performance in Palestinian SMEs. The study recommends that SME owners and managers prioritize employee engagement in addressing safety and sustainability matters to achieve long-term business growth.

Keywords: safety culture; risk mitigation; sustainability practice; business performance; employee engagement

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1. Introduction

In the dynamic business landscape, organizational sustainability is a critical focus for small and medium-sized enterprises (SMEs) worldwide. Particularly in developing economies such as Palestine, SMEs face the dual challenge of maintaining competitive performance while navigating high operational risks, including political instability, financial uncertainty, and resource scarcity. These challenges underscore the importance of integrating safety culture, risk mitigation strategies, and sustainability practices as key drivers of long-term success and resilience for SMEs [1].

The study examines the interrelationship and mutual influence of safety culture, risk mitigation, and sustainability practice in enhancing the resilience and performance of SMEs, particularly in volatile environments like Palestine. These three variables collectively create a synergistic framework for addressing business challenges, where each element reinforces and is reinforced by the others, enabling SMEs to achieve sustainable performance in dynamic and uncertain contexts.

Safety culture forms the foundation of this interplay, fostering shared values, attitudes, and behaviours that prioritize safety across all organizational levels [2]. By proactively identifying and resolving workplace hazards, safety culture reduces incidents and strengthens employee trust and engagement. This alignment with corporate social responsibility (CSR) principles further supports sustainability by promoting ethical practices, enhancing organizational reputation, and minimizing costs related to workplace accidents and regulatory compliance [2,3].

Risk mitigation is the connective tissue between safety culture and sustainability, linking the proactive management of risks to long-term organizational goals. Organizations with strong safety cultures are likelier to adopt structured risk mitigation strategies, such as regular hazard assessments and employee training programmes. These practices reduce financial losses and operational disruptions while aligning with sustainability objectives. For instance, effective risk mitigation enables optimal resource allocation, prevents environmental damage, and protects social well-being, forming a critical pathway to achieving sustainability [4,5].

Sustainability, encompassing economic, environmental, and social dimensions, is influenced by and reinforces safety culture and risk mitigation [2]. Economic sustainability ensures long-term financial stability and operational efficiency, while social sustainability fosters employee well-being, ethical labour practices, and community engagement [6]. Environmental sustainability, in turn, emphasizes responsible resource use and ecological protection [7]. A strong safety culture enhances employee morale, retention, and productivity, essential for sustaining business operations and achieving social and economic stability [8]. Simultaneously, effective risk management reduces workplace hazards, operational disruptions, and environmental damage, contributing to overall sustainability [4]. Integrating these elements creates a reinforcing cycle where sustainability strengthens safety culture and risk mitigation efforts, enhancing organizational resilience and providing competitive advantages [9]. Future research and industry practices should further explore sector-specific examples and best practices to solidify this linkage [10,11].

Empirical evidence highlights the mutual influence among these variables. Studies by Thompson et al. (2021) [6] reveal that organizations with robust safety cultures experience up to a 30% reduction in workplace incidents and significant improvements in employee retention. In comparison, effective risk mitigation strategies enhance operational efficiency by 25% and facilitate sustainability practices. This interconnectedness enables SMEs to adapt to volatile environments and align with global sustainability standards, underscoring the necessity of integrating safety culture, risk mitigation, and sustainability for long-term success [12,13].

Recent research emphasizes the significance of sustainability and risk mitigation in driving business performance [14]. Initiatives like energy efficiency and waste management enhance sustainability, particularly in resource-constrained environments [15]. In volatile contexts, sustainability-oriented business models enable SMEs to balance economic objectives with social and environmental goals, fostering resilience [16]. In Palestine, where SMEs operate under continuous external pressures, strategies that mitigate risks while promoting sustainable development are essential. Contingency planning, risk assessment, and proactive management are vital for SMEs to withstand uncertainties and reduce potential losses [17,18]. Additionally, safety practices related to employee health and safety contribute to better operational efficiency and risk reduction [19].

Central to this study is the role of employee engagement. High levels of engagement are linked to improved productivity, innovation, and overall business performance. Engaged employees are more likely to align with organizational goals, participate in risk management, and contribute to sustainability efforts, thus enhancing the effectiveness of

these initiatives. Studies confirm that organizations adopting safety-conscious and sustainable practices while fostering employee engagement enjoy improved business performance due to increased employee loyalty and reduced turnover [20–22].

This research is grounded in the Resource-Based View (RBV) and Stakeholder Theory. According to RBV, a firm's internal resources, such as engaged employees, offer a sustainable competitive advantage. A culture of shared responsibility, particularly in safety and sustainability initiatives, enhances the effectiveness of these practices. Stakeholder Theory emphasizes the importance of addressing the needs of key stakeholders, including employees, through sustainable practices to ensure long-term business success [23,24].

While much of the existing literature examines sustainability in large corporations or stable economies, a gap exists in understanding how SMEs in volatile regions like Palestine integrate safety culture, risk mitigation, and sustainability practices. Moreover, the mediating role of employee engagement in these relationships remains underexplored, especially in developing economies where employee commitment is crucial for overcoming resource limitations and hostile environments [25].

This study seeks to fill these gaps by examining how safety culture, risk mitigation, and sustainability practices affect the business performance of Palestinian SMEs, with a particular focus on the mediating role of employee engagement. The study will also explore how employee engagement can enhance the impact of these practices on resilience and business growth. This study tries to fill these gaps by providing empirical evidence on how safety, risk mitigation, and sustainable practices influence the business performance of Palestinian SMEs while examining the role of employee engagement as a mediator. The research questions are thus stated below:

1. How do safety culture, risk mitigation, and sustainable practices affect the business performance of Palestinian SMEs?
2. How does employee engagement mediate the relationship between safety, risk mitigation, sustainable practices, and SME performance?
3. How can Palestinian SMEs leverage employee engagement to enhance their resilience and achieve sustainable business growth?

This study provides context-specific insights into how SMEs in Palestine can enhance their resilience and long-term growth by adopting safety, risk mitigation, and sustainability practices. The findings will offer valuable guidance for policymakers, business leaders, and scholars to improve SME outcomes in similar volatile environments. Focusing on SMEs in Palestine, this research develops context-specific insights that can inform policymakers, business leaders, and scholars seeking to enhance SME resilience with sustainable and risk-conscious strategies. The findings will show how an engaged employee can enhance the positive effects of safety and sustainability practices, thus providing a roadmap for SMEs in similar contexts to improve their business outcomes and long-term resilience.

2. Literature Review

2.1. Theoretical Basis

2.1.1. Resource-Based View (RBV)

The RBV emphasizes that businesses can achieve sustained competitive advantage by leveraging intangible resources that are valuable, rare, inimitable, and non-substitutable (Barney, 1991) [24]. In this context, safety culture, sustainability practices, and risk mitigation align with such intangible resources, enabling firms to enhance their competitiveness and long-term success. A safety culture constitutes shared values, attitudes, and practices prioritizing workplace safety. This intangible resource provides a stable and productive work environment, reduces incidents, increases employee satisfaction, and enhances organizational efficiency factors for competitive advantage [26]. Similarly, risk mitigation

represents the unique capability of a firm in identifying, assessing, and managing uncertainties. As an internal resource, the efficiency of risk management reduces disruptions to operations, ensures business continuity, and enhances organizational resilience, especially within turbulent or uncertain markets [27].

In addition, sustainability practices are also strategic intangible assets in terms of aligning an organization's operations with stakeholders' expectations in terms of environmental, social, and economic initiatives. In light of this, sustainable practices let firms attain better efficiency of resources, enhance innovation, and uphold a good reputation to ensure long-term competitive advantages [28]. Among such intangible resources are safety culture, risk mitigation, and sustainability practices that confer resilience, innovation, and operational stability on organizations, particularly those operating within complex and uncertain business environments.

2.1.2. Stakeholder Theory

Stakeholder Theory posits that organizations must consider the interests and expectations of all stakeholders, including employees, customers, investors, suppliers, and the community, to achieve sustainable success [29]. In this context, sustainability practices and risk mitigation are critical tools for addressing stakeholder needs, fostering trust, and ensuring long-term organizational performance. Sustainability practices align with stakeholder expectations by promoting environmental, social, and economic responsibility [20,25]. Today, modern stakeholders want firms to operate ethically, minimize their environmental impact, and take care of societal well-being. Adopting sustainable practices such as reducing carbon footprints, improving employee welfare, and social initiatives will help organizations improve their reputation, stakeholder trust, and long-term support, strengthening their competitive advantage [30].

Similarly, risk mitigation is pivotal in meeting stakeholder expectations by ensuring operational stability and protecting stakeholder interests. Effective risk management practices help organizations anticipate, assess, and respond to potential disruptions, safeguarding business continuity, financial stability, and stakeholder investments [23]. Proactive risk mitigation reassures stakeholders of the organization's resilience and enhances stakeholder confidence in the firm's ability to deliver consistent performance even in uncertain environments [31,32]. By linking sustainability and risk mitigation to stakeholders' expectations, firms take responsibility, engender confidence, and establish long-term relationships with these stakeholders. This corresponds to the demands of their stakeholders and adds value to the organization for continued success and competitiveness in dynamic markets.

2.2. Empirical Literature Review and Hypothesis Development

2.2.1. Safety Culture and Business Performance

Safety culture refers to an organization's collective values, beliefs, attitudes, and practices that prioritize workplace safety. It encompasses the shared commitment to safety that shapes behaviours and decision-making, influencing operational efficiency and the level of risk exposure in the organization [33]. It encompasses critical dimensions such as leadership commitment, employee training, risk awareness, and open communication regarding safety concerns. These dimensions collectively foster an environment where safety is deeply embedded as a core value, influencing employee well-being and organizational outcomes. Safety culture is the intangible organizational resource that boosts organizational performance and resilience in fragile and uncertain economies such as Palestine [24].

A good safety culture essentially helps employees in that there is a secured working environment in which safety is a priority. It minimizes workplace injuries and creates trust, developing job satisfaction, commitment, and loyalty in resource-constrained settings like

Palestinian SMEs, where workers are at the heart of their work; making safety practices a priority earns them trust and security [34]. It heightens productivity, teamwork, and commitment for increased efficiency and stability in the workplace. From an organizational perspective, safety culture decreases disruptions to operations due to workplace accidents, compensation costs, and financial losses from injury [35]. Incorporating safety practices into daily operations translates into better firm continuity, productivity, and economic stability [36]. This is important to SMEs in volatile markets, such as Palestine, characterized by political instability and scarce resources, in light of the purpose of business survivability and competitiveness. In addition, good safety risk management underpins operational resilience, a significant element in long-term sustainability within dynamic environments [32]

Empirical studies consistently establish the critical linkage between safety culture and business performance. The study by Wu et al. (2023) [37] explored the notion that organizations with strong safety cultures have better financial and operational performance. Safety practices in SMEs operating under hazardous conditions lead to improved organizational resilience, operational efficiency, and business concerns [38]. Similarly, Syed-Yahya, Idris, and Noblet (2022) [39] established that a positive safety climate significantly enhances employee motivation, job performance, and organizational performance. A meta-analysis by Kabiesz (2024) [40] further confirmed that safety culture positively influences safety performance, financial stability, and organizational efficiency through reduced disruptions caused by accidents. Research in SMEs from emerging economies further underlines the importance of safety culture to improve workforce morale, retention, and operational stability [41].

While these studies are enlightening, there is still scant empirical research on the role of safety culture in fragile economies such as Palestine. In Palestine, SMEs face unique challenges ranging from political instability to resource scarcity and market uncertainties, heightening the demand for safety culture as an imperative strategic resource [39]. Palestinian SMEs can reduce operational risks by encouraging safety-conscious workplaces and improving organizational resilience, employee stability, and business performance [33]. Despite the increasing evidence on the benefits of safety culture, few studies have explored its role within SMEs in politically fragile economies.

Indeed, Palestinian SMEs face profound uncertainties that impede operations and put their survival at risk. This paper explores how developing a safety culture can be an essential enabler of resilience, which results in better business performance on a sustainable basis due to reduced operational risks within Palestinian SMEs.

Therefore, the hypothesis has been established as follows:

Hypothesis (H1). *Safety culture positively influences business performance in SMEs.*

2.2.2. Risk Mitigation and Business Performance

Risk mitigation is the systematic identification, assessment, and management of potential risks to minimize their impact on organizational performance [42]. Risk mitigation practices are crucial for ensuring operational stability, resource efficiency, and business continuity in uncertain environments such as Palestinian SMEs. These strategies, including proactive planning and risk monitoring, enable firms to adapt to financial, operational, and strategic risks, contributing to long-term success.

Risk mitigation practices have been proven to enhance corporate performance by minimizing disruptions to business operations and promoting efficiency in business processes. For example, Ref. [43] documented that SMEs with structured risk management frameworks exhibit higher operational efficiencies and sound financial performance. Similarly, Shaheen et al. (2020) [44] argued that risk mitigation strategies reduce cash flow volatility,

enabling firms to sustain their profitability and achieve sustainable growth. Research in emerging economies has identified that SMEs with strong risk management systems show greater resilience against exogenous shocks and enhance operational performance [45]. This adaptability enables companies to seize opportunities, even when the environment is unstable; this is particularly important for SMEs in fragile economies like Palestine. For SMEs with few resources, risk mitigation is key to ensuring business continuity in cases of unexpected disruptions.

Efficient risk management ensures better resource allocation, avoidance of unexpected costs, stability of operations, and maintenance of competitiveness [43]. Besides the benefits that operational and financial risks bring, mitigation strategies will affect employee engagement and productivity. A stable and secure work environment, brought about by effective risk management, lets employees build trust and confidence [44]. Employees of organizations with good risk mitigation strategies are more likely to feel safe and valued and, hence, more engaged and satisfied with their jobs. Research by Mansour et al. (2022) [46] corroborates this, stating that employees who perceive their organization to be well prepared for risks are more likely to be engaged, leading to a highly productive workforce that is less likely to leave. Given the political instability, economic hurdles, and market disruptions facing Palestinian SMEs, the need for risk mitigation is an added advantage and one of the survival needs [45].

However, there is a significant gap in research regarding how SMEs in fragile economies like Palestine implement risk management practices. Most existing studies focus on larger corporations or stable markets, leaving the unique challenges of SMEs in unstable regions underexplored. Furthermore, while risk mitigation's operational and financial benefits are well documented by Ghosh et al. (2021) [47], its impact on employee engagement and organizational trust remains under-examined. This gap calls for increased research on how mitigating practices in Palestinian SMEs add to operational success and employee morale for long-term business growth. These will go a long way in helping the current understanding of how risk management in fragile economies affects financial outcomes and workforce dynamics and provide a clearer view of how political and economic instability in Palestine influences approaches to risk mitigation strategies that shape business success.

Hypothesis (H2). *Risk mitigation practices positively influence business success in Palestinian SMEs.*

2.2.3. Sustainability Practice and Business Performance

Sustainability is a business issue of paramount importance, as today's firms are increasingly held responsible for environmental, social, and economic impacts [48]. Sustainability generally involves resource-saving strategies, such as waste reduction and fair labour practices, which are critical to long-term viability. Sustainability can drive differentiation, brand reputation enhancement, and response to increased stakeholder demand for responsible practices in SMEs [46]. Sustainability initiatives often improve operational efficiency and reduce costs, improving financial performance. For example, clean technologies can cut costs while boosting productivity. The orientation of firms to sustainability enhances competitiveness and diminishes risks due to resource scarcity and regulations [49]. Eco-innovations decreased costs and improved quality and market share, especially for SMEs operating in the most challenging markets with lean approaches [50].

It also allows for the establishment of a strong brand reputation and further enriches the competitive advantage. Organizations embracing sustainability tend to be associated with responsibility and ethics; in this respect, it improves stakeholder interaction with customers, suppliers, and investors. In times when consumers show an increasing inclination

for companies and brands that share values with theirs, sustainability practices would place an organization differently from others in the market [51]. A study by Flammer in 2013 [52] indicated that those firms that integrated sustainability into their operations were more likely to attract loyal customers and investors who valued environmental and social responsibility. Sustainability initiatives may drive innovation, enabling SMEs to adapt to shifting consumer preferences and develop new products or services that conform to environmental and social trends [48]. Indeed, many research works establish the positive link between sustainability and business performance. For example, studies of developed markets indicate that those companies that show strong sustainability practices outperform others in financial measures like ROI and profitability [53].

In the context of developing economies, integrating sustainability into core strategies has yielded higher innovation, competitiveness, and long-term growth among SMEs [54]. However, a gap exists in reviewing how sustainability impacts SMEs within fragile economies like Palestine. Palestinian SMEs' specific challenges due to political instability, a low resource base, and market volatility indicate that sustainability may affect their performance differently. While the positive effect of sustainability on business performance is well documented within the global literature, in Palestine, this research context remains limited. Few studies have discussed how sustainability initiatives might influence driving operational efficiency, financial success, and competitive advantage in Palestinian SMEs.

Hypothesis (H3). *Sustainability practices positively influence business performance in Palestinian SMEs.*

2.3. Employee Engagement as a Mediator

2.3.1. Employee Engagement and Business Performance

Employee engagement refers to employees' emotional commitment to their organization, influencing their performance and organizational outcomes. On the other hand, business performance measures a company's success, usually reflected in financial results, profitability, and overall sustainability. This literature review explores theoretical and empirical studies examining the relationship between employee engagement and business performance (Macey & Schneider, 2008) [55]. Therefore, to implement the study, several theoretical frameworks were proposed to elucidate the connection between employee engagement and business performance.

One such framework is the Resource-Based View (RBV), which suggests a firm can achieve a competitive advantage by leveraging its human capital, including engaged employees [56]. Engaged employees, as valuable resources, are instrumental in achieving superior business outcomes. When employees are engaged, they are more likely to invest effort in their tasks, leading to enhanced organizational performance.

In business performance, employee engagement is a mediator that influences how various organizational practices and leadership behaviours contribute to performance outcomes [57]. However, limited research explicitly addresses employee engagement's direct and indirect effects on business performance, and studies on this relationship remain inconclusive. Several researchers argue that employee engagement is a critical driver of organizational success, with engaged employees contributing to better customer service, higher innovation, and more effective problem-solving, all of which can lead to improved business performance [58]. Hence, this study proposes that employee engagement mediates the relationship between organizational practices (e.g., leadership, culture) and business performance. By fostering engagement, companies may see improved business outcomes, such as increased profitability and sustained growth [59].

Hypothesis (H4). *Employee engagement positively impacts business performance.*

2.3.2. Safety Culture, Employee Engagement and Business Performance

Within the landscape of Palestinian SMEs, a nuanced blend of challenges and opportunities shapes the interactions between safety culture, employee engagement, and business performance [60]. A positive safety culture, characterized by shared values and practices regarding workplace safety, establishes a foundation for a secure and efficient work environment in these enterprises [61]. Employee engagement emerges as a pivotal factor in Palestinian SMEs, influencing how employees perceive and respond to safety initiatives and shaping the overall safety culture within organizations.

However, empirical research by Liu et al. (2022) [62] shows a positive correlation: a strong safety culture exists that is positively associated with a higher degree of engagement established among the employees. Employees in a company may show more commitment, motivation, and pro-action toward safety practices if the safety culture is the primary focus of building safety consciousness.

Furthermore, the impact of employee engagement on safety performance extends into broader business performance measures for SMEs themselves [62]. Therefore, engaged employees are more productive, innovative, and customer-oriented, adding to the competitiveness and sustainability of such enterprises within the economically challenging environment by acknowledging the mediating role of employee engagement in the relationship between safety culture and business performance in Palestinian SMEs. Organizations can then develop tailored strategies to enhance engagement levels, cultivate positive safety practices, and drive comprehensive performance improvements [63].

Therefore, the interlinked dynamics of safety culture, employee engagement, and business performance make understanding and leveraging these factors important for Palestinian SMEs. With targeted interventions and strategic research studies, organizations will be able to release the full potential of their workforce, improve safety, and work towards attaining sustainable business performance in the SME sector in Palestine. From the perspective of the specific contextual features of SMEs in Palestine, the following hypotheses were developed concerning the association of the safety culture and employee engagement with business performance:

Hypothesis (H5). *Employee engagement significantly mediates the relationship between safety culture and business performance within Palestinian SMEs.*

2.3.3. Risk Mitigation, Employee Engagement and Business Performance

Risk mitigation strategies are essential for businesses aiming to reduce uncertainty and enhance operational effectiveness. Effective risk mitigation safeguards a company's assets and fosters a work environment where employees are more engaged, contributing to improved business performance. The Resource-Based View (RBV) suggests that companies implementing strong risk management practices develop unique capabilities that protect them from external shocks and internal inefficiencies, leading to higher employee engagement and sustained performance [24].

Employee engagement is crucial in the relationship between risk mitigation and business performance. When employees perceive that their organization effectively mitigates risks, they experience less job-related stress and greater job satisfaction [64]. The study creates a positive feedback loop, where engaged employees are more productive, innovative, and committed, ultimately driving business performance [65]. On the other hand, sound risk management makes employees feel safe in the work environment. In return, such employees will pay back through increased engagement. The positive consequences will be better teamwork, reduced turnover, and enabling organizational culture—all aspects that contribute to improved business performance [66].

Nevertheless, the size and industry of the organization may be influencing factors that vary concerning how risk mitigation impacts employee engagement and business performance. High-risk sectors, such as construction or manufacturing, will probably suffer more impact since mitigating risks impinge on employee safety and welfare [64]. Therefore, effective risk mitigation improves employee engagement, mediating between risk management and business performance for increased productivity and better organizational outcomes.

Hypothesis (H6). *Employee engagement significantly mediates the relationship between risk mitigation and business performance.*

2.3.4. Sustainability Practice, Employee Engagement, and Business Performance

Employee engagement is an essential channelling factor in the association between sustainability practices and business performance [67]. While sustainability practices lay the foundation for organizations to minimize risks and improve performance, their proper execution depends on employee involvement. Engaged employees have been instrumental in successfully applying sustainability initiatives through planning, execution, and continuous improvement of sustainability practices [68]. Accordingly, employee engagement is an essential bridge between sustainability practices and actual improvements in business performance.

Some studies have identified employee engagement as a key driver of business performance through sustainability practices. For instance, Farmanesh et al. (2023) [67] established that employee engagement mediates between leadership behaviours and organizational outcomes, such as sustainability practices and business performance. In Palestinian SMEs, employee engagement ensures the inculcation of sustainability practices into the business's daily operations, which leads to improved efficiency, cost savings, and better market positioning [69].

Moreover, engaged employees guarantee the long-term success and sustainability of adapted practices. Active employees support the sustainability initiatives in the company and drive innovative ideas by proposing new ideas for improving current practices. These continuous innovation cycles keep Palestinian SMEs competitive in the market; they can adapt to demands in the market, which affects business performance [70].

Hypothesis (H7). *Employee engagement mediates the relationship between sustainability practices and business performance in Palestinian SMEs.*

This paper, therefore, proposes a model using employee engagement as a mediator to test the effects of safety culture, risk mitigation, and sustainability on business performance in Palestinian SMEs. The framework postulates that SMEs can achieve higher performance by developing a sound safety culture and adopting measures for mitigating risks, which enhance employee engagement. In turn, highly engaged employees increase productivity and innovation, leading to business success. In addition, the impact of risk mitigation and sustainability practices on business performance is mediated by employee engagement. Through sustainable practices, firms enhance their reputation and attract stakeholders to improve performance. These relations may be very different concerning company size, industry, and availability of resources in Palestinian SMEs. The model incorporates employee engagement as a mediator, business performance as the dependent variable, and safety culture, risk mitigation, and sustainability as independent variables. Figure 1 describes the conceptual model showing the relationship between these variables.

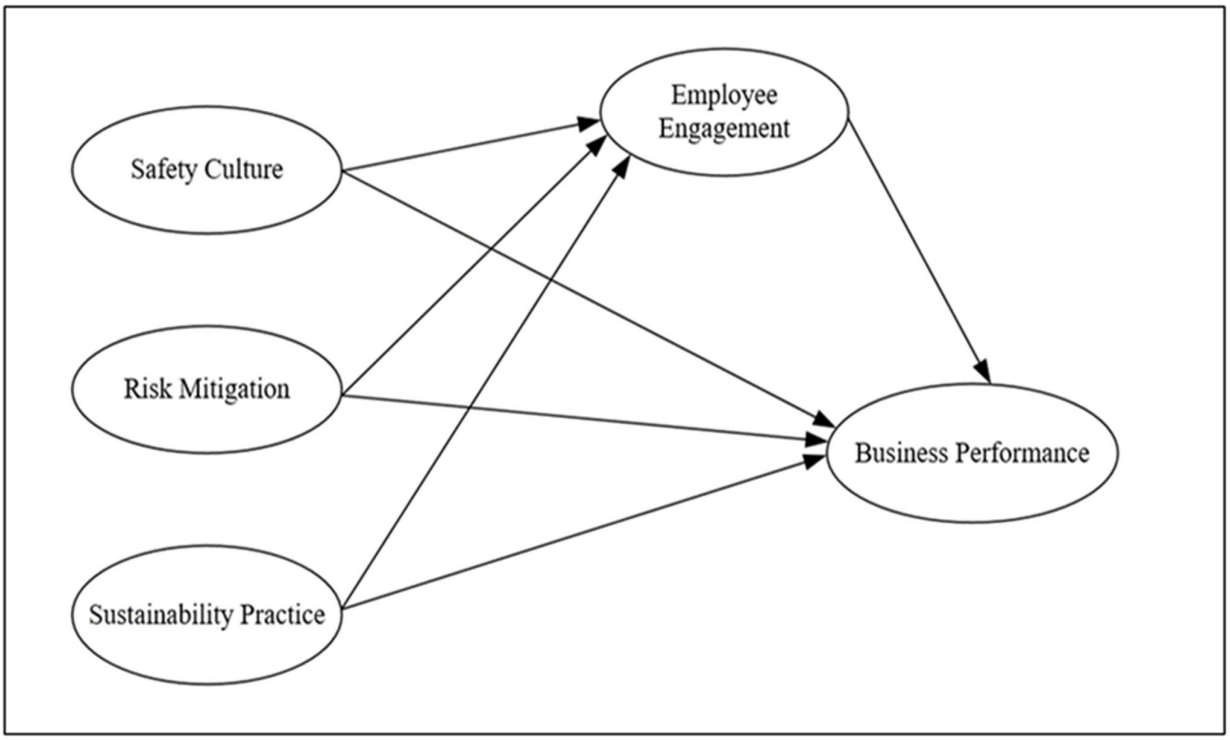


Figure 1. Authors' proposed conceptual model for the study.

3. Research Design

3.1. Variable Operationalization

3.1.1. Independent Variables

Safety Culture

This study conceptualized safety culture as an independent variable of safety awareness, organizational practices, and management commitment. Safety awareness was measured by employees' knowledge and engagement with safety procedures. Organizational practices focused on adherence to safety protocols and reporting mechanisms. Management commitment was assessed by leadership involvement in fostering safety. The measurement items examined safety integration in the organizational culture, including safety training, risk assessments, and industry compliance. These dimensions highlight safety culture's impact on employee engagement, incident reduction, and operational efficiency, promoting employee well-being and sustainable practices, adapted from Neal and Griffin (2006) and Singer et al. (2009) [71,72]. All the measurement items are found in the Appendix A.

Risk Mitigation

Risk mitigation refers to strategies and mechanisms that help identify, assess, and reduce business risks. For this study, the risk mitigation indicators were adapted from Aven, T. (2007) [73], which include proactive risk assessment, the adoption of contingency plans, investment in risk management technologies, and regular training in risk mitigation strategies. These indicators also examine the effectiveness of risk communication, risk monitoring systems, and the capacity for recovery from unexpected disruptions. In the present study, a 5-point Likert scale was used to assess the extent of the implementation of these practices and their impact on operational resilience and continuity. All the measurement items are found in the Appendix A.

Sustainability Practice

The sustainability practice assessment will consider how far environmental, social, and governance initiatives have been embedded in organizational operations. Sustainability practices in this study are measured based on the indicators adapted from Renwick, Redman, and Maguire (2013) [74]. Measurement items include resource conservation initiatives, waste reduction, stakeholder engagement in sustainability programmes, and adherence to global sustainability standards. These indicators assess an organization's initiatives for reducing carbon footprints, ethics-based supply chain management, and employees' involvement in an organization's sustainable performance. In measuring the degree of adoption of sustainability practices in enhancing long-term business performance and societal impact, a 5-point Likert scale was used. All the measurement items are found in the Appendix A.

3.1.2. Dependent Variable

Business Performance

Business performance measures critical financial and non-financial indicators that reflect the organizations' operational efficiency, profitability, and market competitiveness. The business performance indicators in this study have been adapted from Hopkin (2018) [75]. The measurement items include financial performance metrics such as return on investment, revenue growth, and profit margin, as well as non-financial metrics such as customer satisfaction, employee productivity, and market share. The indicators will also capture the effectiveness of strategic initiatives, innovation adoption, and resource utilization. Business performance is assessed on a 5-point Likert scale, focusing on the ability of the organization to achieve sustainable growth, enhance stakeholder value, and maintain competitive advantage in the marketplace. All the measurement items are found in the Appendix A.

3.1.3. Mediating Variable

Employee Engagement

The mediating variable will be employee engagement. Observed variables will measure employee engagement as an indirect way of measuring the influence of employee engagement on organizational outcomes relating to business performance, safety culture, and sustainability practices; measurement items for the mediating variable were adapted from Mata-Lima et al. (2016) [76]. The indicators will be vigour, dedication, and absorption, showing how much employees are emotionally and cognitively invested in their work. In measuring the degree or level of employee engagement in productivity, innovation, and organizational alignment, a 5-point Likert scale was applied to the ratings. All the measurement items are found in the Appendix A.

3.2. Data Collection and Sampling

The research focuses on how safety culture, risk mitigation, and sustainability practices influence business performance in Palestinian SMEs, considering employee engagement as the mediating variable. This research discusses how an organization integrates appropriate safety protocols, risk mitigation strategies, and sustainable practices to achieve business success. In other words, the critical concerns are the level of safety awareness, the effectiveness of mitigation strategies to handle risks, and the application of sustainability practices aligned with the operational goals. The sample is selected using a purposive sampling method, targeting Palestinian SMEs across various sectors that prioritize safety, risk management, and sustainability in their operations. This method is appropriate as it ensures the inclusion of directly relevant SMEs to the research objectives, allowing for focused data collection from entities with experience and practices in these areas. The study

aims to generate meaningful insights into the relationship between safety culture, risk mitigation, and sustainable business performance by selecting participants with specific characteristics and expertise. The study employs deductive reasoning and quantitative research methods [77] to analyze the relationships among the variables and their collective impact on organizational outcomes.

The study delved into how safety culture, risk mitigation, and sustainability practices influence the performance of Palestinian SMEs, with a focus on the mediating role of employee engagement. A purposive sample of 720 SMEs from manufacturing, service, and agricultural sectors, explicitly targeting 450 SMEs with advanced safety, risk management, and sustainability practices, was selected from the study area of the West Bank (Ramallah, Hebron, Nablus, and Jericho). In the study of deductive reasoning and quantitative methods, primary data were gathered from owners who are responsible for the business.

Respondents were strategically chosen using purposive sampling to ensure comprehensive insights, emphasizing senior management and employee perspectives on safety and sustainability measures. A self-administered questionnaire was distributed, with 22 incomplete responses excluded from the analysis. The remaining reactions exceeded 10% of the SME population, meeting the recommended thresholds for sample size validity. The study strived for representativeness by including SMEs from diverse regions in Palestine, ensuring size and sector representation variations. The research design and sample selection process were methodical and aimed to provide a robust understanding of the relationship between safety culture, risk mitigation, sustainability practices, employee engagement, and business performance in Palestinian SMEs. To ensure representativeness, the sample size exceeded 10% of the population, adhering to the recommended threshold [78]. To enhance readability and ensure transparency, offering a concise presentation of findings, the sample selection process is outlined in Table 1.

Table 1. The sample selection process in tabular form.

Step	Description	Criteria/Procedure
1	Total Population	SMEs operating in Palestine across manufacturing, service, and agricultural sectors.
2	Sample Frame	SMEs that actively implement safety culture, risk management strategies, and sustainability practices.
3	Sampling Technique	The purposive sampling method is based on specific criteria, including safety culture initiatives, risk mitigation measures, and sustainability practices.
4	Sample Size Determination	Determined based on practical considerations and available resources. There is a sample of 720 SMEs, with 450 chosen explicitly for their advanced safety and sustainability efforts.
5	Sample Selection	Selection of SMEs based on the defined criteria and accessibility of data. Purposive sampling ensures relevant representation across different sectors.
6	Data Collection	Primary data were collected through self-administered questionnaires distributed to business owners, senior managers, and employees, and secondary data were collected from reports.
7	Data Analysis	PLS-SEM analysis using the Smart-PLS, version 4, statistical tool to examine the relationships among constructs and conduct mediation analysis.

3.3. Data Analysis

This study also performed a PLS-SEM analysis using the Smart-PLS statistical tool to examine the relationships between the constructs. PLS-SEM is a second-generation regression model with several functions and two distinct components [79]. A measurement model was employed in the analysis's initial section to rate the dependability and accuracy of many indicators. The current inquiry employed the structural model to support the hypothesis and show the structural linkages between the variables. The study's mediation analysis also used PLS-SEM to determine the construct's direct and indirect effects [80].

4. Results

4.1. Descriptive Statistics

The study involved 428 respondents from Palestinian SMEs in the manufacturing and service sectors. Demographics showed a gender imbalance (67.8% male, 32.2% female) and an experienced workforce (majority aged 40–49 with over a decade of experience). Education levels were high (doctorate: 8.9%; master's 29.4%; bachelor's: 17.3%). Roles varied (employees: 21.0%; supervisors: 23.4%; managers: 28.1%; business owners: 27.6%), reflecting a diverse, educated, and experienced group. These insights from Table 2 show the sample's capability to provide valuable information on safety, risk mitigation, and sustainability practices in Palestinian SMEs. Table 2 suggests that most respondents could comprehend and contribute helpful information for the study.

Table 2. Demographic information.

Demographic Variables	Details	Frequency	%
Geographic Location			
Business Characteristics	SMEs	428	100
Industry Sectors	Manufacturing and Services		
Gender			
Male		290	67.8%
Female		138	32.2%
Age	20–29, 30–39, 40–49, ≥50		
20–29		30	7.0%
30–39		100	23.4%
40–49		150	35.1%
≥50		148	34.6%
Education			
Doctorate		38	8.9%
Masters		126	29.4%
Bachelor		74	17.3%
Experience (Years)	Years of experience in current role		
5–10		70	16.4%
11–15		120	28.1%
16–20		140	32.7%
>20		98	22.9%
Role in the Company	Employees, Supervisors, Managers, Business Owners		
Employees		90	21.0%
Supervisors		100	23.4%
Managers		120	28.1%
Business Owners		118	27.6%

4.2. Measurement Model Analysis

The measurement model is part of the model that examines the relationship between latent variables and their measures [71]. We used it to prove the constructs' validity and reliability. The measurement model in PLS-SEM serves as the starting point for evaluating the accuracy and dependability of the indicators.

4.2.1. Reliability and Validity Test

PLS-SEM statistical techniques were used to evaluate the reliability and validity of each relevant questionnaire item. The measuring model assessed the validity and reliability using factor loadings, Cronbach's alpha, composite reliability, convergent validity, and discriminant validity [81]. The study's convergent validity results are thus displayed in Table 3 below. Cronbach's alpha and composite dependability are used to gauge the scale items' internal consistency [82]. Cronbach's alpha and composite reliability should, on average, be more than 0.70, while average variance extracted (AVE) and factor loadings should, on average, be more significant than 0.50 [83]. As shown in Table 3 and Figure 2, which confirm that the sample data in the current study are robust and reliable, the Cronbach alpha and C.R. values are higher than 0.70. The results also show that the AVE and factor loading values are more significant than 0.50, indicating substantial correlations between the items and actual convergent validity.

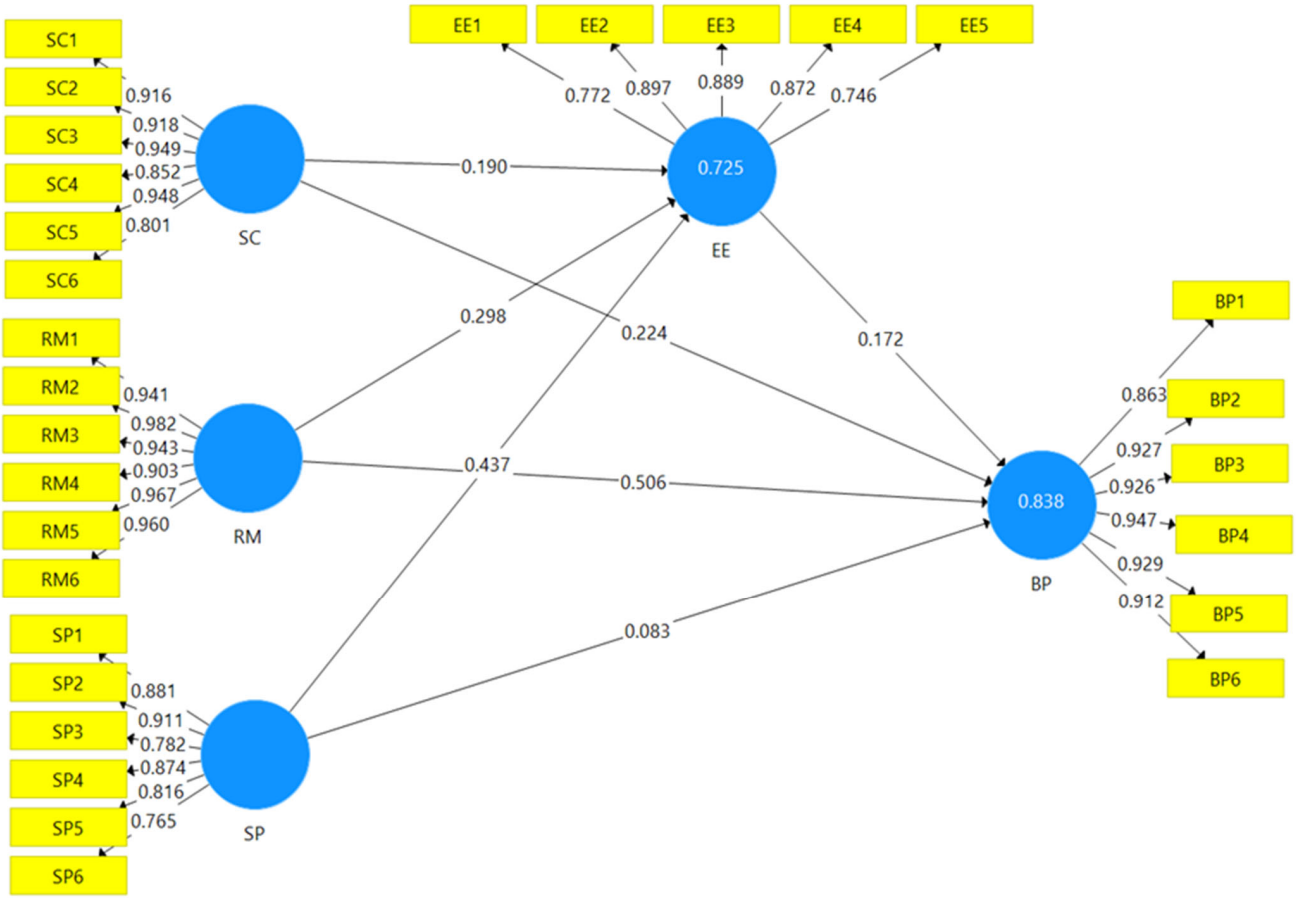


Figure 2. Measurement model assessment (output).

Table 3. Measurement model analysis.

Constructs	Items	F. Loading	Cronbach's Alpha	Composite Reliability	(AVE)
Safety Culture	SC1	0.916	0.952	0.962	0.808
	SC2	0.918			
	SC3	0.949			
	SC4	0.852			
	SC5	0.948			
	SC6	0.801			
Risk Mitigation	RM1	0.941	0.978	0.982	0.902
	RM2	0.982			
	RM3	0.943			
	RM4	0.903			
	RM5	0.967			
	RM6	0.96			
Sustainability Practice	SP1	0.881	0.916	0.935	0.705
	SP2	0.911			
	SP3	0.782			
	SP4	0.874			
	SP5	0.816			
	SP6	0.765			
Employee Engagement	EE1	0.772	0.896	0.921	0.702
	EE2	0.897			
	EE3	0.889			
	EE4	0.872			
	EE5	0.746			
Business Performance	BP1	0.863	0.962	0.97	0.842
	BP2	0.927			
	BP3	0.926			
	BP4	0.947			
	BP5	0.929			
	BP6	0.912			

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

4.2.2. Discriminant Validity

Using statistics, we first evaluated the convergence and reliability before examining the discriminant validity. The goal was to demonstrate no data redundancy when using items that assess the same construct [84,85]. Three techniques were applied to establish discriminant validity in the Smart-PLS: cross-loadings, the Heterotrait–Monotrait (HTMT) ratio analysis, and the Fornell–Larcker criterion [86]. First, we looked at the discriminant validity using the correlation (Larcker, 1981) [87]. The criterion states that discriminant validity is guaranteed if the square root of the AVE for a specific concept is more significant than its correlation with other constructs [88]. The squared AVE value in the diagonal of Table 4 indicates no problem with discriminant validity because it is bigger than the inter-construct correlation, shows that a construct is distinct, and explains notions not represented by other constructs in the model [89].

Table 4. Fornell–Larcker criterion.

	BP	EE	RM	SC	SP
BP	0.918				
EE	0.803	0.838			
RM	0.888	0.779	0.95		
SC	0.838	0.76	0.834	0.899	
SP	0.758	0.797	0.738	0.736	0.84

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

The second step used CL (cross-loading) to verify the discriminant validity. Per this method, a specific item should have more significant parent constructions than other constructs in the investigation [87]. According to Table 5, the item has higher loadings than the others in the current investigation, showing a weak link between the variables and comprehensive discriminant validity.

Table 5. Cross-loadings.

	BP	EE	RM	SC	SP
BP1	0.863	0.664	0.728	0.686	0.646
BP2	0.927	0.734	0.818	0.777	0.687
BP3	0.926	0.727	0.82	0.768	0.685
BP4	0.947	0.762	0.836	0.78	0.713
BP5	0.929	0.761	0.835	0.789	0.711
BP6	0.912	0.764	0.844	0.806	0.728
EE1	0.495	0.772	0.448	0.467	0.546
EE2	0.701	0.897	0.668	0.638	0.683
EE3	0.781	0.889	0.785	0.746	0.735
EE4	0.809	0.872	0.801	0.786	0.774
EE5	0.465	0.746	0.433	0.437	0.536
RM1	0.846	0.758	0.941	0.804	0.731
RM2	0.859	0.752	0.982	0.814	0.714
RM3	0.847	0.742	0.943	0.79	0.691
RM4	0.819	0.713	0.903	0.781	0.65
RM5	0.855	0.741	0.967	0.793	0.707
RM6	0.833	0.731	0.96	0.769	0.709
SC1	0.823	0.77	0.82	0.916	0.73
SC2	0.8	0.743	0.798	0.918	0.721
SC3	0.783	0.715	0.774	0.949	0.674
SC4	0.665	0.583	0.674	0.852	0.581
SC5	0.782	0.709	0.773	0.948	0.673
SC6	0.638	0.544	0.636	0.801	0.568
SP1	0.709	0.723	0.687	0.676	0.881
SP2	0.761	0.759	0.755	0.717	0.911
SP3	0.551	0.582	0.515	0.511	0.782
SP4	0.706	0.753	0.715	0.703	0.874
SP5	0.546	0.593	0.507	0.545	0.816
SP6	0.487	0.56	0.467	0.507	0.765

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

The Heterotrait–Mononitrate (HTMT) correlation ratio is used as a final test of discriminant validity. A threshold level of 0.90 is acceptable if the route model comprises constructs that are conceptually quite similar. This study the HTMT value is found in the range ratio of 0.774–0.87. In the principle of HTMT, if the HTMT value is higher than 0.90, it denotes a lack of discriminant validity. Hence, HTMT values in our case were less than

0.90, demonstrating the variables' low correlations and adequate discriminant validity [90]. The results are shown in Table 6 below.

Table 6. Heterotrait–Monotrait ratio.

	BP	EE	RM	SC	SP
BP					
EE	0.833				
RM	0.813	0.797			
SC	0.87	0.786	0.86		
SP	0.795	0.853	0.765	0.774	

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance

4.3. Structural Model Analysis

4.3.1. Assessment of Collinearity Issues

Assessing multi-collinearity is a crucial step in structural model analysis, as it ensures the reliability of the results. This study evaluated collinearity using the Variance Inflation Factor (VIF). The VIF values, ranging from 1.694 to 2.674 (see Table 7), are all below the threshold of 3, indicating no multi-collinearity concerns. The highest VIF value of 2.674, between safety culture (SC) and employee engagement (EE), remains well within the acceptable range [91]. Thus, the collinearity in the model is deemed sufficient.

Table 7. Collinearity statistics (VIF).

	BP	EE	RM	SC	SP
BP					
EE	2.636				
RM	2.018	1.694			
SC	1.805	2.574			
SP	2.147	2.453			

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

4.3.2. Evaluation of Path Coefficients

The purpose of evaluating path coefficients is to examine the strength, significance, and direction of relationships between variables. Bootstrapping (5000 resamples) calculates the path coefficients and corresponding *p*-values. The importance of each relationship is assessed, with positive relationships indicated by $\beta > 0$.

In Table 8, the path coefficients show that all relationships are significant ($p < 0.01$). For instance, the path from safety culture (SC) positively influences business performance (BP) ($\beta = 0.257$, $p < 0.01$), supporting H1. Risk mitigation (RM) and business performance (BP) have a strong positive relationship ($\beta = 0.557$, $p < 0.01$) (H2), supporting the hypothesis that risk mitigation significantly influences business performance. Similarly, the path from sustainability practice (SP) to BP ($\beta = 0.158$, $p < 0.01$) and the path from employee engagement (EE) to BP ($\beta = 0.172$, $p < 0.01$) are both significant, suggesting that sustainability practices and employee engagement also positively influence business performance; H3 and H4 are supported.

Table 8. Path analysis results.

Paths	Original Sample (O)	Mean (M)	(STDEV)	T Statistics	*p*-Value
EE → BP	0.172	0.174	0.047	3.682	0.000
RM → BP	0.557	0.557	0.047	11.971	0.000
RM → EE	0.298	0.301	0.075	3.99	0.000
SC → BP	0.257	0.256	0.044	5.796	0.000
SC → EE	0.19	0.19	0.064	2.95	0.003
SP → BP	0.158	0.159	0.04	3.983	0.000
SP → EE	0.437	0.436	0.056	7.734	0.000

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

Additionally, safety culture (SC) positively influences both employee engagement (EE) (β = 0.19, $p < 0.01$) and business performance (BP) (β = 0.257, $p < 0.01$), supporting H1. Risk mitigation (RM) significantly influences employee engagement (EE) (β = 0.298, $p < 0.01$). Sustainability practice (SP) also has a significant effect on both employee engagement (EE) (β = 0.437, $p < 0.01$) and business performance (BP) (β = 0.158, $p < 0.01$); H5, H6, and H7 are supported. The path analysis results support the hypotheses and demonstrate significant, positive relationships between the constructs, reinforcing the model's robustness. In addition, the structural model (results of path analysis) is shown in Figure 3.

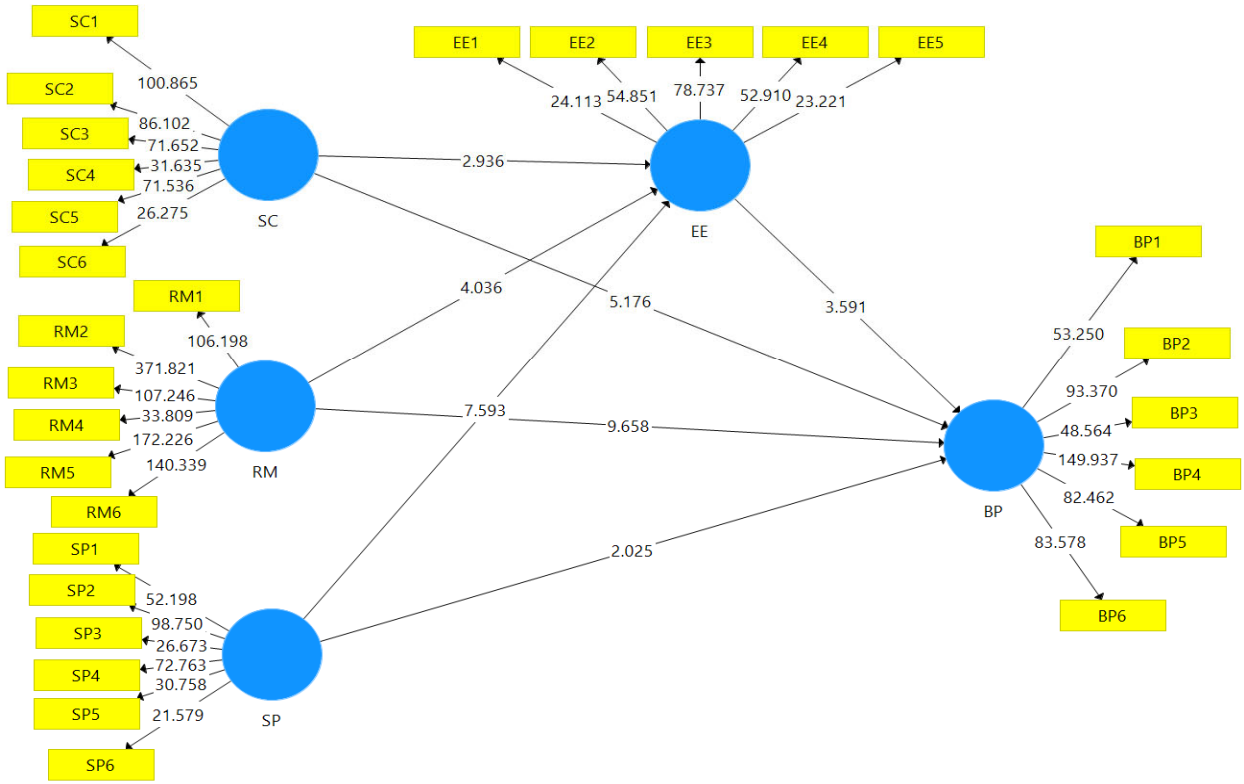


Figure 3. Structural model (results of path analysis).

4.3.3. Coefficient of Determination (R^2)

The Coefficient of Determination (R^2) measures the proportion of variance in the dependent variables that the predictors in the model explain. R^2 values for endogenous constructs are reported to assess the model's explanatory power, with the magnitude of R^2 indicating the strength of the relationships (weak, moderate, or substantial). As shown in Table 9, the R^2 value for employee engagement is 0.73, meaning that the predictors

explain 73% of the variance in employee engagement. This represents a substantial level of explanation. Similarly, the R^2 value for business performance is 0.84, indicating that the model explains 84% of the variance in business performance, reflecting a high degree of explanatory power.

Table 9. Coefficient of Determination (R^2).

Explained Variance for Each Dependent Variable (R^2)	
Employee Engagement	**Business Performance**
73%	84%

4.3.4. Effect Size ($f2$)

Effect size (f2) quantifies the contribution of individual predictors to the R^2 of dependent variables. f2 values are classified as small (f2 > 0.02), medium (f2 > 0.15), or large (f2 > 0.35). As shown in Table 10, risk mitigation (RM) exhibits a large effect size on business performance (BP) (f2 = 0.394), indicating a strong impact. Employee engagement (EE) has a medium effect size on BP (f2 = 0.154), while sustainability practice (SP) shows a medium effect on EE (f2 = 0.323). Other relationships, such as safety culture (SC) and BP (f2 = 0.231), also indicate moderate contributions, reflecting the substantial influence of these predictors on the model.

Table 10. Effect size ($f2$).

	BP	**EE**	**RM**	**SC**	**SP**
BP					
EE	0.154				
RM	0.394	0.247			
SC	0.231	0.126			
SP	0.194	0.323			

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

4.3.5. Mediation Analysis

The mediation analysis, conducted using bootstrapping with 5000 resamples, as summarized in Table 11, reveals that employee engagement (EE) partially mediates the relationships between the independent variables (risk mitigation, safety culture, and sustainability practice) and business performance (BP). Specifically, the indirect effect of risk mitigation (RM) → EE → BP was significant ($\beta = 0.051$, $p = 0.03$), indicating partial mediation. Similarly, the safety culture (SC) → EE → BP path also demonstrated partial mediation with a significant effect ($\beta = 0.033$, $p = 0.016$). The most substantial indirect effect was observed in the sustainability practice (SP) → EE → BP path ($\beta = 0.075$, $p < 0.001$), further supporting employee engagement as a mediator. All mediation effects were significant based on the bootstrapping results, confirming that employee engagement significantly and partially mediates the impact of these predictors on business performance.

Table 11. Mediation effects of employee engagement.

Paths	**Coefficient**	**SD**	**T-Value**	***p*-Value**	**Significance**
RM → EE → BP	0.051	0.022	2.368	0.03	Partial
SC → EE → BP	0.033	0.015	2.21	0.016	Partial
SP → EE → BP	0.075	0.02	3.852	0.000	Partial

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

4.3.6. Goodness-of-Fit Measures

The goodness-of-fit measures for the structural model, as summarized in Table 12, indicate a satisfactory model fit. The Standardized Root Mean Square Residual (SRMR) value of 0.072 is below the threshold of 0.08, signifying an acceptable fit between the model and the data. The Unweighted Least Squares Distance (d_ULS) of 0.03 is low, suggesting a minimal discrepancy between observed and model-implied values, indicating a firm fit. Similarly, the Geodesic Distance (d_G) value of 0.005 supports the model's robustness, showing minimal differences between observed and estimated values. While high, the Chi-Square (χ^2) value of 3077.678 is considered acceptable in this context due to its sensitivity to sample size and in light of other favourable fit indices. Lastly, the Normed Fit Index (NFI) of 0.886 exceeds the threshold of 0.80, confirming a good fit. Collectively, these metrics validate that the structural model fits the data well.

Table 12. Goodness-of-fit measures.

	Standard Value	Estimated Model	Acceptance
SMRMR	<0.08	0.072	Fit
d_ULS	low	0.03	Fit
d_G	low	0.005	Fit
Chi-Square (χ^2)	Lower value	3077.678	Fit
NFI	>0.8	0.886	Fit

Note: SRMR = Standardized Root Mean Square Residual; d_ULS = Unweighted Least Squares Distance; d_G = Geodesic Distance; NFI = Normed Fit Index and results from the author's analysis through SEM-PLS.

4.3.7. Hypothesis Testing

The results of the hypothesis testing, as summarized in Table 13, demonstrate significant support for all proposed relationships. Specifically, H1 posited that safety culture (SC) would positively influence business performance (BP), and the analysis confirmed this with a significant coefficient ($\beta = 0.257$, $t = 5.796$, $p < 0.001$). Similarly, H2 hypothesized a positive relationship between risk mitigation (RM) and BP, which was strongly supported ($\beta = 0.557$, $t = 11.971$, $p < 0.001$). H3 tested the impact of sustainability practice (SP) on BP, and the results again supported a positive and significant effect ($\beta = 0.158$, $t = 3.983$, $p < 0.001$). Furthermore, H4 examined the direct impact of employee engagement (EE) on BP, and this path was also statistically significant ($\beta = 0.172$, $t = 3.682$, $p < 0.001$), confirming the hypothesis.

Table 13. Summarized results testing the hypotheses.

Hypotheses	Paths	Coefficient	T-Value	*p*-Value	Decision
H1(+)	SC → BP	0.257	5.796	0.000	Supported
H2(+)	RM → BP	0.557	11.971	0.000	Supported
H3(+)	SP → BP	0.158	3.983	0.000	Supported
H4(+)	EE → BP	0.172	3.682	0.000	Supported
	Mediation Effects of Employee Engagement				
H5(+)	SC → EE → BP	0.033	2.21	0.016	Supported
H6(+)	RM → EE → BP	0.051	2.368	0.03	Supported
H7(+)	SP → EE → BP	0.075	3.852	0.000	Supported

Note: SC = safety culture; RM = risk mitigation; SP = sustainability practice; EE = employee engagement; BP = business performance.

In terms of mediation, H5 explored the indirect effect of SC on BP through EE, revealing a significant partial mediation ($\beta = 0.033$, $t = 2.21$, $p = 0.016$). H6 tested the indirect

relationship between RM and BP via EE, and the results supported partial mediation (β = 0.051, t = 2.368, p = 0.03). Lastly, H7 demonstrated that the mediation of SP on BP through EE was the most robust, with a significant coefficient (β = 0.075, t = 3.852, $p < 0.001$). These findings confirm that employee engagement significantly mediates the relationships between the independent variables and business performance. In conclusion, the hypotheses were strongly supported, highlighting the significant direct and indirect effects of safety culture, risk mitigation, and sustainability practices and the mediating role of employee engagement in enhancing business performance.

5. Discussion and Implications for Policymakers

5.1. Discussion

The findings of this study provide robust evidence of the critical roles of safety culture, risk mitigation, and sustainability practices in driving business performance within Palestinian SMEs. These results underscore the importance of these practices in fostering competitive and resilient SMEs and address several gaps in the existing literature by highlighting the mediating influence of employee engagement. This study extends previous findings by demonstrating how an integrated approach to organizational practices and employee involvement can create a synergistic effect that promotes sustainable growth.

5.1.1. Impact of Safety Culture on Business Performance

This study confirms the significant and positive relationship between safety culture and business performance, echoing findings from previous research [92]. However, it moves beyond the existing literature by contextualizing the relationship within Palestinian SMEs, a largely underexplored and unique environment. Unlike prior studies examining safety culture in large enterprises or developed economies, this research provides nuanced insights into how safety culture functions in resource-constrained SMEs operating in volatile economic and socio-political settings.

Specifically, the findings highlight the dual role of safety culture as a compliance-driven necessity and a strategic enabler of operational excellence [2]. While previous studies have often emphasized safety compliance as a regulatory obligation, this research underscores its potential to enhance employee productivity, morale, and overall organizational outcomes when adopted strategically. Safety leadership, in particular, emerges as a crucial factor in shaping a culture of accountability and trust, driving safer work environments and fostering innovation and agility, key components of sustainable business performance.

Additionally, this study enriches the discourse by linking safety culture to resilience and adaptability, traits critical for SMEs navigating unpredictable economic conditions. Unlike large corporations, SMEs often lack the financial and operational buffers against weather crises. This research shows that a robust safety culture can mitigate such vulnerabilities by fostering a proactive mindset and collaborative behaviours among employees, enabling SMEs to respond more effectively to external shocks.

Compared to earlier studies, this research provides a more comprehensive understanding by integrating organizational resilience and adaptability into the safety culture–business performance relationship. The existing literature has often overlooked these dimensions, focusing primarily on safety outcomes, such as accident rates or compliance metrics. This study demonstrates that safety culture can catalyze broader organizational improvements, including enhanced employee engagement, operational efficiency, and long-term sustainability.

Furthermore, the context of Palestinian SMEs sheds light on the challenges and opportunities associated with implementing safety practices in environments characterized by limited resources and heightened risks. While prior studies in developed economies may assume stable institutional frameworks, this research reveals how safety culture can

stabilize in less predictable settings. By providing empirical evidence from a region where SMEs face unique constraints, the study adds depth to the existing body of knowledge. It underscores the importance of tailoring safety strategies to specific organizational and environmental contexts.

In conclusion, this study not only corroborates prior findings on the positive impact of safety culture but also extends the literature by emphasizing its role in fostering resilience, adaptability, and sustainable performance in SMEs. These insights contribute to a more holistic understanding of safety culture's strategic importance, particularly in resource-limited and high-risk settings like Palestinian SMEs.

5.1.2. Impact of Risk Mitigation on Business Performance

This study affirms the significant impact of risk mitigation practices on business performance, aligning with existing research while offering a novel perspective by focusing on SMEs in high-risk environments like Palestine. Unlike studies centred on large organizations in stable economies, this research emphasizes how tailored risk management strategies empower SMEs to navigate disruptions effectively while identifying opportunities for innovation and market growth [4,90]. The findings underscore the critical role of comprehensive risk assessments, which enhance organizational adaptability and resilience when integrated with active employee engagement. This integration enables SMEs to proactively address external uncertainties, maintain operational stability, and sustain competitive advantage.

Moreover, this study bridges a key gap in the literature by situating risk mitigation within the context of resource-constrained SMEs in volatile economic settings. It highlights how strategic risk management fosters risk avoidance and strategic agility, aligning with broader organizational goals. By expanding the theoretical understanding of risk management, particularly in developing economies, this research provides actionable insights into how SMEs can leverage risk mitigation practices to achieve sustained business performance [70]. In conclusion, this study enriches the discourse by illustrating the dual benefits of risk mitigation, reducing vulnerabilities and driving growth in challenging environments

5.1.3. Impact of Sustainability Practice on Business Performance

This study highlights the transformative potential of sustainability practices in driving long-term business success, particularly within Palestinian SMEs. While previous research [88] has predominantly focused on the advantages of sustainability in large corporations, this study broadens the scope by demonstrating how SMEs, even in resource-constrained settings, can harness sustainable practices to achieve significant business outcomes. The findings reveal that integrating sustainability's environmental, social, and economic dimensions enables SMEs to enhance operational efficiency, foster stakeholder trust, and build customer loyalty [93]. For instance, environmentally conscious practices reduce operational costs through resource efficiency and appeal to environmentally aware consumers, boosting brand loyalty [94]. Similarly, social sustainability initiatives, such as employee welfare programmes and community engagement, strengthen internal and external stakeholder relationships, fostering a sense of organizational purpose and trust.

Moreover, the study addresses a critical gap by linking sustainability practices directly to strategic agility, underscoring their role in enhancing SMEs' resilience to market volatility. In a volatile economic environment such as Palestine, the ability to adapt to changing market dynamics and leverage sustainability for competitive advantage becomes a crucial determinant of survival and growth. Hence, this study enriches the existing literature by demonstrating the multifaceted benefits of sustainability practices for SMEs, emphasizing

their role in fostering operational efficiency, stakeholder trust, and strategic adaptability in challenging contexts.

5.1.4. Mediating Role of Employee Engagement

This study makes a distinctive contribution by demonstrating the mediating role of employee engagement in linking organizational practices to business performance. While Naji et al. (2021) [89] emphasized the pivotal role of employee engagement in fostering organizational success, this research extends their findings by revealing how engagement serves as a critical mechanism for translating safety culture, risk mitigation, and sustainability practices into measurable performance improvements.

Specifically, the findings show that engaged employees in Palestinian SMEs are more innovative, collaborative, and dedicated, amplifying the impact of organizational practices on overall performance. For example, safety culture initiatives become more effective when employees actively foster safe work environments, while risk mitigation efforts gain traction through heightened employee awareness and proactive problem-solving. Similarly, sustainability practices are better implemented and sustained when employees feel invested in the organization's goals and values.

This study addresses a notable gap in the literature by uncovering how employee engagement operationalizes the link between organizational practices and performance, particularly in resource-constrained SMEs. These insights offer actionable strategies for SMEs to enhance workforce involvement, emphasizing the need for management to prioritize engagement as a strategic lever for driving business success.

5.2. Implications

5.2.1. Theoretical Implication

This study presents essential theoretical insights into the relationships between safety culture, risk mitigation, sustainability, and employee engagement in driving business success among Palestinian SMEs. It established a significant positive impact of safety culture, risk mitigation, and sustainability practices on business performance, thus emphasizing the value of holistic risk management strategies in enhancing organizational resilience and competitiveness. These findings support prior research on the role of proactive risk management in boosting performance [95]. The mediating role of employee engagement underlines its crucial function in translating safety practices, risk mitigation, and sustainability efforts into complex business outcomes. An engaged employee is one of the most significant driving forces linking organizational strategies to performance improvement, proving the motivating and commitment logic in the workforce.

The research findings indicate that organizational elements are interrelated, demonstrating a synergistic relationship between safety culture, risk mitigation, sustainability, and employee engagement [96]. This integrated approach aligns with the Resource-Based View (RBV), emphasizing that leveraging internal resources—such as safety practices and employee engagement—provides a pathway to achieving sustainable competitive advantages [4]. These insights enrich the existing literature on organizational behaviour and performance. They offer significant practical implications for practitioners and policymakers, particularly within SME contexts like Palestine, as they strive to enhance sustainability and success.

Moreover, this research paves the way for future studies to explore additional factors influencing business survival in transitional regions, such as the role of cultural and institutional support. By shedding light on these dynamics, the study contributes valuable theoretical insights to the limited literature on SMEs in Palestine, particularly regarding the factors driving their growth and development.

5.2.2. Practical Implication

The empirical findings of this study provide some functional, practical implications for Palestinian SMEs in the use of safety culture, risk mitigation, sustainability, and employee engagement to attain business sustainability. It is indicated that developing a safety culture can be advantageous for Palestinian SMEs in terms of minimizing operational hazards and improving workplace safety. These companies can create this culture by emphasizing safety practices, frequent safety audits, comprehensive training, and employee involvement in safety matters [97].

This study also underlines the contribution of risk mitigation practices in achieving business performance. Moreover, Palestinian SMEs could reinforce risk management by developing efficient processes for assessing risk, preparing contingency plans, and using insurance to mitigate potential losses. Integrating these approaches will make them more resilient and ensure continuity should unexpected setbacks occur. The research also draws on the importance of sustainability practices that drive long-term success. By adopting sustainable practices such as energy efficiency, waste reduction, and community engagement, SMEs can minimize their environmental impact, enhance their reputation, and attract socially conscious customers [98]. Initiatives such as green technologies, recycling programmes, and corporate social responsibility activities will place SMEs in a different light as responsible citizens.

In addition, employee engagement is the most essential part of business success. To enhance engagement, SMEs should create open communication, provide opportunities for skill development, recognize employee contributions, and promote a positive work culture. An engaged workforce is more productive, creative, and loyal, improving performance and sustainability. The research provides practical insights for enhancing safety, mitigating risks, embracing sustainability, creating employee engagement, and ensuring long-term success for Palestinian SMEs.

6. Conclusions

This study advances the understanding of how safety culture, risk mitigation, and sustainability practices contribute to business performance in Palestinian SMEs, emphasizing the mediating role of employee engagement. Addressing research gaps and incorporating recent findings provides a nuanced perspective on the interplay of organizational practices and employee dynamics in achieving sustainable growth. These insights lay a robust foundation for future research and practical interventions to foster competitive and resilient SMEs in developing economies.

Despite its contributions, this study has certain limitations. The cross-sectional design restricts causal inferences, and future research could adopt longitudinal methodologies to capture the dynamic interactions between variables over time. Additionally, the focus on Palestinian SMEs limits the generalizability of the findings. Comparative studies across diverse regions and industries could provide a more comprehensive understanding of how safety, risk, and sustainability practices drive business performance in varying SME contexts. Further exploration of moderating variables, such as industry type or organizational size, could also enrich the analysis. Lastly, we acknowledge that the use of purposive sampling, a non-probability sampling technique, represents a limitation of this study.

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Appendix A

Table A1. Measurement construct items.

Variables	Items	Source
Safety Culture	(1) Management demonstrates a visible and consistent commitment to ensuring safety in the workplace. (2) Employees actively participate in safety initiatives and contribute to improving safety processes. (3) Safety-related information is openly shared across all levels of the organization. (4) All employees receive regular and comprehensive safety training tailored to their job roles. (5) Employees feel safe and encouraged to report safety issues without fear of retaliation. (6) The organization proactively identifies and mitigates potential safety hazards before incidents occur.	[71,72]
Risk Mitigation	(1) The organization systematically identifies potential risks before they impact operations. (2) All identified risks are thoroughly analyzed to assess their likelihood and potential impact. (3) The organization implements proactive measures to prevent risks from occurring. (4) Risk mitigation strategies are regularly reviewed and adjusted to address emerging risks. (5) Employees receive training on risk mitigation practices relevant to their roles. (6) The organization has effective contingency plans to manage unexpected risk-related crises.	[73]
Sustainability Practice	(1) The organization implements strategies to reduce environmental impacts, such as energy and resource efficiency. (2) The organization engages in activities that promote the well-being of local communities. (3) The organization ensures long-term financial stability through sustainable business practices. (4) The organization regularly publishes reports on its sustainability performance and goals. (5) Employees are actively encouraged to participate in sustainability-related initiatives. (6) The organization prioritizes suppliers and partners who adhere to sustainable practices.	[74]
Employee Engagement	(1) I feel a strong emotional connection to my work and the organization. (2) Enthusiastic and dedicated to fulfilling my job responsibilities. (3) Deeply focused and fully absorbed in my work activities. (4) Encouraged to share my ideas and opinions to improve work processes. (5) My organization values my contributions and supports my well-being.	[75]

Table A1. *Cont.*

Variables	Items	Source
Business Performance	(1) The organization has achieved significant growth in revenue, profitability, and return on investment. (2) The organization effectively utilizes resources to improve operational efficiency and minimize costs. (3) The organization is competitive by offering superior products and services. (4) The organization consistently meets or exceeds customer expectations in terms of product quality and customer service. (5) The organization continuously introduces new products, services, or processes that contribute to its growth and market leadership. (6) The organization fosters a work environment that enhances employee performance and engagement, leading to overall business success.	[76]

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